Connecting the A3 printers

Before connecting the printers make sure you have:

- Read *How to connect the Raspberry PI* (Control Engineering)
- You have installed SPERTE successfully.

List of Connections: There are doubles as some of these components connect with each other. This is a list that contains each necessary connection for each component.

- **Raspberry Pi**
  - USB-C power cable (top outlet).
  - USB-C to laptop (bottom outlet).
  - Ethernet cable to Ebox.
- **Ebox**
  - Power cable.
  - Ethernet cable to Raspberry PI.
  - Digital in and out from A3 printer.
  - Encoder 1 from printer.
  - BNC cable from analog out to amplifier +/− 2.5 V.
- **Amplifier**
  - Power cable.
  - +/− 2.5 V in to Ebox Analog Out 1.
  - Load from printer.

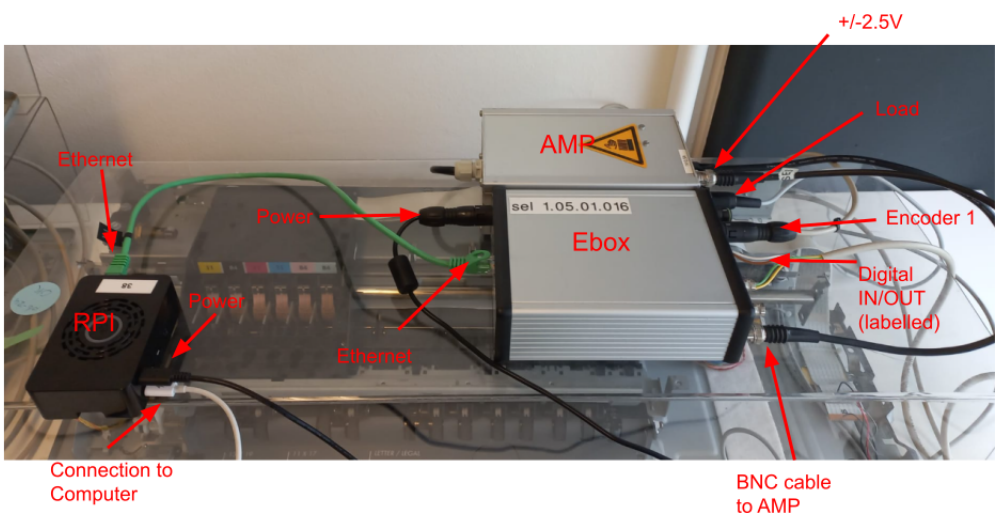

Figure 1: Schematic of Connections.

Procedure

1. Connect all the cables and turn power on.
2. Open the MATLAB experimental script (provided or your own).
3. Open the Simulink model of the printer.
4. Click hardware settings --> Hardware implementation --> Target Hardware Resources --> Board parameters --> fill in the following:
   (a) Device address: 10.55.0.1
   (b) Username: pi
   (c) Password: sperte123
5. Running the files
   (a) Initialize the model.
   (b) Click on monitor & tune.
   (c) Run the MATLAB script.
   (d) If the feedforward signal is above 2.5 V refer to `Good to know` before applying the signal.
6. When finished with the experiments run `SPERTE Shutdown`

Good to know

- After running `SPERTE Shutdown` wait 10 seconds before turning power off.
- The PI is slow to connect.
- Spikes above 2.5 V in the feedforward signal are possible and these are due by the connection of the Raspberry PI and Ebox. If possible stop the experiments and restart everything to let the PI and amplifier cool down. If the spike is necessarily too high the feedforward can still be applied but beware that the printer might break (accelerate uncontrollably and crash on the airbag).
- Make sure ALL the cables are connected.
- Always make your reference a numeric vector, there must be no NaN. If this is the case, the printer will accelerate uncontrollably and crash.